Review guide

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Short, and ordered by how much your judgement is needed rather than by how the paper is organised.

The four things most worth your scepticism

1. The frame orientation (section on the frame orientation)

The congruence between the split $(5,5)$ oscillator metric and the Lorentzian $(1,9)$ metric is solved modulo p via a square root, which has two values differing by an orientation reversal. Orientation flips the Hodge star, which exchanges the eigenspace the gamma map annihilates.

This was found by a holdout prime failing, misdiagnosed as a $p \equiv 3 \pmod 8$ rule on two examples, and then refuted when 32633 — which is 1 mod 8 — turned out to need the same branch. **If the pinning convention is not the one you would choose, the bridge section changes.**

2. $\dim_{\mathbb{Q}} D_{10} = 11$

D_{10} is the activated flow closure. The first attempt at this computation took the raw span of all degree-ten flow targets, got 14, and would have reported a quotient of 0 — no obstruction at all. The distinction between the raw span and the activated closure is the whole result.

The question for you is whether D_{10} as defined is the physically right object, not whether the arithmetic is right; the arithmetic is checked two ways.

3. Rank 81

The count is yours and your coauthors'. What is contributed here is a certificate: an explicit 81×81 minor, nonzero at six primes by two independent determinant routines. **Is that a contribution worth the space the paper gives it?**

4. Degree-eight ablation

Span equality at degree eight holds only with the structured tensor-word family; the port-graph family alone reaches rank six of seven. Reported as a property of the candidate family, not of the bridge. **Is that the right reading?**

What the paper deliberately does not claim

- degree-twelve tensor–spinor equivalence

- any all-order result
- a complete invariant-ring presentation
- canonicality or uniqueness of any basis
- any physical or type IIB consequence of the flow result
- invention of the enumerate–evaluate–relate method

Where the numbers come from

Every number in the paper is generated from a JSON certificate at build time. There are no hand-typed values; the build fails if an artifact is missing. The reproduction quickstart shows how to regenerate any of them.

What would change your mind is worth telling us

If a convention is wrong, the affected section is named in the claim ledger along with what else depends on it.